

Problem 1.

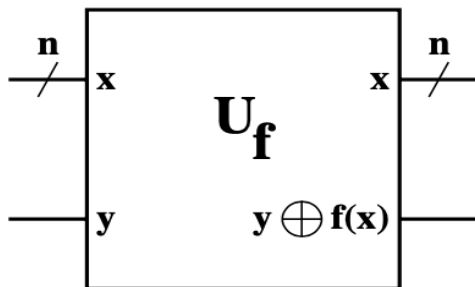
Quantum Circuit for Boolean Functions

Let $f : \{0, 1\} \rightarrow \{0, 1\}$ be a Boolean function which outputs a bit for each input bit, and let $U_f : |x, y\rangle \rightarrow |x, y \oplus f(x)\rangle$.

Show that U_f is unitary.

More generally, let $f : \{0, 1\}^n \rightarrow \{0, 1\}$ be a Boolean function of n bits with $U_f : |x, y\rangle \rightarrow |x, y \oplus f(x)\rangle$. Now U_f has $n + 1$ input qubits (the first n are input to f) and $n + 1$ output qubits (the first n are identical with the first n qubits of the input) as shown in the figure below.

Check that for any integer n the operator U_f is a unitary transformation.



Uncomputing Garbage

As described in class, any classical AND/OR/NOT circuit computing a function

$$f : \{0, 1\}^n \rightarrow \{0, 1\}^m$$

can be efficiently converted to a reversible circuit using only Toffoli gates. The process of making each gate reversible produces some “garbage” outputs that should be erased at the end of the computation.

Call C_f the reversible gate computing f and assume that the input register includes some workspace bits 0^a which produce some garbage bits j^a in output.

Draw a scheme of a circuit for reversibly computing $f(x)$ by erasing the content of the output and the workspace registers. (Hint: use CNOT gates to reversibly copy the result into some extra m 0-bits.)

Problem 2.*The Hadamard Transform*

Show that

$$H^{\otimes n}|0\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle.$$

and

$$\begin{aligned} H|x\rangle &= \sum_{y \in \{0,1\}} (-1)^{xy} |y\rangle / \sqrt{2} \\ H^{\otimes n}|x\rangle &= \sum_{y \in \{0,1\}^n} (-1)^{x \cdot y} |y\rangle / \sqrt{2^n} \end{aligned}$$

1-Qubit Gates

The exponential of the Pauli matrices give rise to three classes of unitary matrices, namely the rotation operators about the x, y and z axes, defined by:

$$R_x(\theta) \equiv e^{-i\theta\sigma_x/2} = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} \sigma_x = \begin{bmatrix} \cos(\theta/2) & -i \sin(\theta/2) \\ -i \sin(\theta/2) & \cos(\theta/2) \end{bmatrix} \quad (1)$$

$$R_y(\theta) \equiv e^{-i\theta\sigma_y/2} = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} \sigma_y = \begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix} \quad (2)$$

$$R_z(\theta) \equiv e^{-i\theta\sigma_z/2} = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} \sigma_z = \begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix}. \quad (3)$$

- (a) Let $x \in \mathbb{R}$ and A a matrix such that $A^2 = I$. As shown in class

$$e^{-iAx} = \cos(x)I + i \sin(x)A.$$

Use this result to verify Equations (1), (2) and (3).

- (b) Show that the Hadamard gate H can be expressed as a product of R_x and R_z and $e^{i\varphi}$ for some φ .

Problem 3. Universality

1. Show that if for a single qubit unitary operation U we have $U = V^2$ where V is another single qubit unitary operation, then the double controlled U gate, called $C^2(U)$ gate, can be implemented as in Figure 1. Show that for $V = (1 - i)(I + iX)/2$ this implements the Toffoli gate.

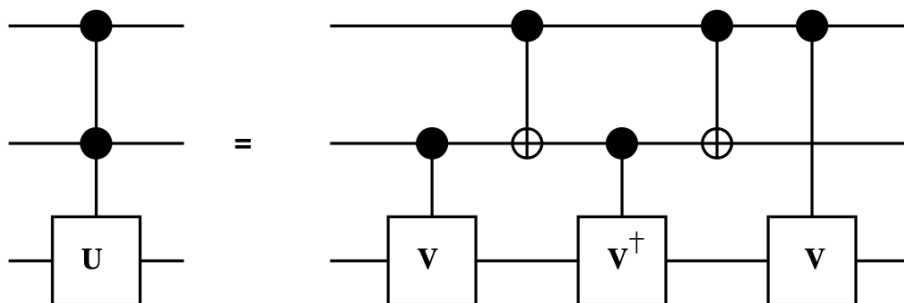


Figure 1: Implementation of $C^2(U)$

2. Check that $XYX = -Y$ and $XZX = -Z$; then show that $XR_y(\theta)X = R_y(-\theta)$ and $XR_z(\theta)X = R_z(-\theta)$.
3. Assuming that the single qubit operation U is given by $U = e^{i\alpha}AXBXC$ with $ABC = I$, show that the network in Figure 2 implements the controlled U gate, $C(U)$, using the single qubit gate X the phase gate α and the single qubit gates A , B and C .

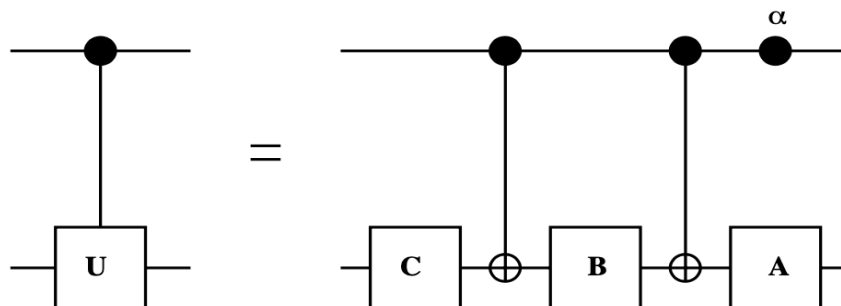


Figure 2: Implementation of $C(U)$